

Scientific Computation

Prof. Matthew Giamou
Sept. 23, 2025

Lecture 10 - LU Decomposition

Lecture Outline

- Solving $Ax = b^{(i)}$ repeatedly

Computing m times repeatedly

- Elementary matrix operations
- The LU decomposition

Motivation

Suppose we have to solve the system $Ax = b^{(i)}$ for $i = 1, \dots, m$, which often occurs in scientific computation

Can we do better than performing Gaussian elimination m times, which gives an approach that is $O(mn^3)$?

↳ Find a way to solve $Ax = b^{(i)}$ faster

First Idea: Inversion

First idea: Invert A (i.e., use Gaussian elimination to find A^{-1}) and use this in subsequent calculations

$$x = A^{-1}b = \begin{bmatrix} -a_{1n} & \dots & -a_{11} \\ \vdots & & \vdots \\ -a_{nn} & \dots & -a_{n1} \end{bmatrix} \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}, \quad x_k = \sum_{i=1}^n a_{ki} b_i$$

$\rightarrow$ elements of A^{-1}
 n multiplications
 n additions

But, computing A^{-1} takes more than just GE on $[A|b]$

we solve for A^{-1} by GE on $[A|I] \rightarrow O(n^3)$
 (Gauss-Jordan)

Note: $O(n^3 + mn^2) < O(mn^3)$
 LU Before

Second Idea: Decomposition

Second idea: Backwards substitution is the fast part of direct solutions. What if we could efficiently **store** the results of Gaussian elimination in a way that is better than A^{-1} ?

LU decomposition.

$A = LU \rightarrow$ the result of recording steps for GE as matrix operations.

L is lower triangular matrix

U is upper triangular matrix

Then solve $Ax = LUx = b$

$$\begin{aligned} z &= Ux \\ \therefore Lz &= b \end{aligned}$$

Lecture Outline

- Solving $Ax = b^{(i)}$ repeatedly
- **Elementary matrix operations**
- The LU decomposition

Row Subtraction

How can we write the row-subtraction in Gaussian elimination as a linear transformation (i.e., a matrix)?

$$A \left\{ \begin{bmatrix} \text{---} a_1^T \text{---} \\ \text{---} a_2^T \text{---} \\ \text{---} \vdots \text{---} \end{bmatrix} \right\} \mapsto \begin{bmatrix} \text{---} a_1^T \text{---} \\ a_2^T - l_{2,1} a_1^T \\ \text{---} \vdots \text{---} \end{bmatrix}$$

$$MA = \begin{pmatrix} 1 & & & 0 \\ & \ddots & & \\ 0 & & 1 & \end{pmatrix} \begin{bmatrix} -a_1^T \\ \vdots \\ -a_n^T \end{bmatrix} = A + \begin{bmatrix} 0 \\ -b_{21}a_1^T \\ \vdots \\ 0 \end{bmatrix}$$

7

Row Swapping

What about row swapping for partial pivoting?

$$\begin{bmatrix} -a_1^T \\ -a_2^T \\ \vdots \\ -a_n^T \end{bmatrix} \mapsto \begin{bmatrix} -a_2^T \\ -a_1^T \\ \vdots \\ -a_n^T \end{bmatrix}$$

$$\text{Ex. } \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \mapsto \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

This effect is equivalent to a permutation matrix

- A row swap (permutation)
matrix is its own inverse

↳ 2 each row = 2 each col
= 1,
 $a_{ij} \in \{0, 1\}$

8

GE In Matrix Form

GE IN MATRIX FORM

In summary, Gaussian elimination has the following matrix structure:

$$\underbrace{M_k \cdots M_2 M_1}_{A^k} x = \underbrace{M_k \cdots M_2 M_1}_{b^k} b \quad [A | b]$$

$M_1 \downarrow$

$$[A' | b']$$

$\vdots$

$\downarrow M_k$

$$[A^k | b^k]$$

Each step of GE is multiplication of $[A | b]$ by an elementary matrix

M_k :

$$[A^k | b^k] = M_k [A^{k-1} | b^{k-1}]$$

LU Structure

This should enable our idea to efficiently store the results of GE:

$(M_{n-1} \cdots M_1) A = A^{n-1} \rightarrow$ result of GE, so it's in REF. and therefore upper triangular.

$$(M_{n-1} M_{n-2} \cdots M_1) = \begin{bmatrix} 1 & & & 0 \\ -l_{21} & 1 & & \\ \vdots & & \ddots & \\ -l_{n1} & & & 1 \end{bmatrix} = L^{-1} \text{ (L inverse)}$$

$$L = M_1^{-1} M_2^{-1} \cdots M_{n-1}^{-1}$$

$$M_k = I - l_k e_k^T$$

$$M_k^{-1} = I + l_k e_k^T$$

Lecture Outline

- Solving $Ax = b^{(i)}$ repeatedly
- Elementary matrix operations
- The LU decomposition

A Small Example

Example 1: Consider the matrix

$$A = \begin{bmatrix} 1 & -1 & 3 \\ 1 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix} \begin{array}{l} \downarrow 1 \\ \downarrow 3 \end{array} \rightarrow \begin{array}{l} \text{multiplications} \\ l_{21}=1 \\ l_{31}=3 \end{array}$$

$$M_1 A = \begin{bmatrix} 1 & 0 & 0 \\ -l_{2,1} & 1 & 0 \\ -l_{3,1} & 0 & 1 \end{bmatrix} A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 3 \\ 1 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}$$

↓

First step of GE
in matrix form

$$= \begin{bmatrix} 1 & -1 & 3 \\ 0 & 2 & -3 \\ 0 & 1 & -8 \end{bmatrix}$$

12

A Small Example

Example 1 cont.: Our next multiplier is $l_{3,2} = \frac{1}{2}$, so

$$M_2 A^{(1)} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -\frac{1}{2} & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 & 3 \\ 0 & 2 & -3 \\ 0 & 1 & -8 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 3 \\ 0 & 2 & -3 \\ 0 & 0 & -6.5 \end{bmatrix} = A^{(2)} = U$$

$$M_2 A^{(2)} = M_2 M_1 A = U$$

$$\begin{aligned} \therefore A &= (M_2 M_1)^{-1} U \\ &= M_1^{-1} M_2^{-1} U \\ &= L U \end{aligned}$$

13

A Small Example

Example 1 cont.: We have U , so the lower triangular component is

$$L = M_1^{-1}M_2^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{1}{2} & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & \frac{1}{2} & 1 \end{bmatrix}$$

$$\therefore \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & \frac{1}{2} & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 1 & -1 & 3 \\ 0 & 2 & -3 \\ 0 & 0 & -6.5 \end{bmatrix}}_U = \underbrace{\begin{bmatrix} 1 & -1 & 3 \\ 1 & 1 & 0 \\ 3 & -2 & 1 \end{bmatrix}}_A$$

14

Solving with L & U

Example 1 cont.: Next, we want to solve

$$Lz = b, \text{ where } z = Ux. \quad \xrightarrow{\quad} L(Ux) = b$$

Begin by solving for z :

$$Lz = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 3 & \frac{1}{2} & 1 \end{bmatrix} z = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = b,$$

using **forward substitution** (which is $O(n^2)$) gives

$$z_1 = 1,$$

$$z_2 = 2 - z_1 = 2 - 1 = 1,$$

$$z_3 = 3 - 3z_1 - \frac{1}{2}z_2 = 3 - 3 - \frac{1}{2} = -\frac{1}{2}$$

15

Solving with L & U

Example 1 cont.: Finally,

$$LUx = Lz \implies Ux = z$$

and we have solved for z so we can apply **backward substitution** on

$$Ux = \begin{bmatrix} 1 & -1 & 3 \\ 0 & 2 & -3 \\ 0 & 0 & -6.5 \end{bmatrix} x = \begin{bmatrix} 1 \\ 1 \\ -\frac{1}{2} \end{bmatrix} = z,$$

to get (in $O(n^2)$):

$$x_3 = \frac{1}{13},$$

$$x_2 = 1 - 3x_3 = \frac{1}{2}\left(1 + \frac{3}{13}\right) = \frac{8}{13},$$

$$x_1 = 1 + x_2 - 3x_3 = 1 + \frac{8}{13} - \frac{3}{13} = \frac{18}{13}$$

16

General Description of LU Decomposition

Letting $l_{i,k} = a_{ik}/a_{kk}$ for $i = 2, \dots, n$, step k looks like

$$\begin{bmatrix} 1 & 0 & \dots & \dots \\ 0 & 1 & \dots & \dots \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix} \begin{bmatrix} a_{11}^{(k-1)} & a_{12}^{(k-1)} & \dots & a_{1n}^{(k-1)} \\ 0 & a_{22}^{(k-1)} & a_{23}^{(k-1)} & \dots \\ \vdots & \vdots & \ddots & \vdots \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & \dots & 0 \\ \vdots & 1 & 0 & & \vdots \\ & \vdots & 1 & & \\ & & -l_{k+1,k} & 1 & \\ & & -l_{k+2,k} & 0 & \ddots \\ & & \vdots & \vdots & \\ 0 & 0 & \dots & -l_{n,k} & 0 & 0 & \dots & 1 \end{bmatrix} \begin{bmatrix} \vdots & 0 & a_{33}^{(k-1)} & \dots & a_{3n}^{(k-1)} \\ & \vdots & 0 & \ddots & a_{3n}^{(k-1)} \\ & & \vdots & a_{kk}^{(k-1)} & \dots & a_{kn}^{(k-1)} \\ & & & a_{(k+1)k}^{(k-1)} & \dots & a_{(k+1)n}^{(k-1)} \\ & & & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 0 & a_{nk}^{(k-1)} & a_{nn}^{(k-1)} \end{bmatrix} = A^{(k)}$$

17

General Description of LU Decomposition

In summary,

$$M_{n-1}M_{n-2}\cdots M_2M_1A = A^{(n-1)} \triangleq U$$

which is upper triangular because the M_k matrices carry out Gaussian elimination. Solving for A gives us:

18

简单来说，当我们用GE来求解线性系统的时候，由于前面我们的论证，时间复杂度是 $O(n^3)$ ，但是问题

是，即使我们让A不变，仅仅改变b，所求解x的时间都是 $O(n^3)$ 。所以我们想找一种方法，当虬结过一次之后，如果我们仅仅改变b，系统能以更快的速度求解。

已知我们使用化简好的REF矩阵来forward substitute或者back sub的时间复杂度都是 $O(n^2)$ ，同时，GE的过程可以写作成一堆matrix的相乘，所以我们使用矩阵 L^{-1} 来表示这些矩阵相乘的结果，同时让 $LU=A$ ，U为经历过完整的GE之后得到的upper triangular matrix。这样的好处是我们记录了GE的过程（用 L^{-1} 来表示），所以在第一次计算的时候我们是 $O(n^3)$ ，但是之后的计算都是 $O(n^2)$ 。

这样的话，我们就可以写成 $LUx=b$ ，或者 $Lz=b$ ， $z=Ux$ ，先求解z，这一步是 $O(n^2)$ ，然后求解 $Ux=z$ ，求解x，这一步也是 $O(n^2)$ ，所以我们成功的把时间复杂度从每一次都 $O(n^3)$ 降低到只有第一次 $O(n^3)$ ，后续都是 $O(n^2)$ 。